

Global Memory

MEMORY TRACKER (Jul): ~20% QoQ price increase in 3QCY26



Mark Li
+852 2123 2645
mark.li@bernsteinsg.com



Mark C. Newman
+1 212 845 7822
mark.newman@bernsteinsg.com



Edward Hou, CFA
+852 2123 2623
edward.hou@bernsteinsg.com



Yipin Cai, CFA
+852 2123 2669
yipin.cai@bernsteinsg.com



April Li
+1 917 344 8339
april.li@bernsteinsg.com



Phoebe Sun
+1 917 344 8481
phoebe.sun@bernsteinsg.com

This monthly tracker (download dataset [here](#)) summarizes contract/spot price data released by TrendForce/DRAMeXchange & compares that with our model & investor expectations.

DRAM: Jul contracts suggested 17% conventional DRAM price increase QoQ in 3QCY26.

Spot price: DRAM spot price was fairly strong in Jul. PC DRAM spot price rose by 8.8-17% MoM while server DRAM was mixed at -6.7% to up 3.8% MoM.

Contract price: DRAM contract prices in Jul rose broadly MoM & indicated 3QCY26 contract price to be 17% higher than 2QCY26, with 17% for PC, 16% for Server, 11% for Mobile & ~27% for Consumer. The size of the price increase is a tad ahead of our model, but narrowed notably vs. 2Q and is probably smaller than some of the latest Street expectations. PC & Mobile demand continued softening & customers were very reluctant to accept further price hikes. Server demand was still very robust, as the overall supply situation is expected to become tighter into next year, although the price hike was somewhat attenuated by LTA price caps.

NAND: Wafer contract price was stagnant MoM. Mobile and SSD price increases continued though perhaps not as strong as expected.

Spot price: NAND wafer spot price was mixed with 512Gb falling 3% MoM while 1Tb wafer rebounding 8% MoM.

Contract price: Wafer contract price was largely unchanged MoM in Jul, since module makers were concerned about consumer demand & mostly rejecting further price hikes. Transaction volume was also extremely low. Mobile NAND price increase was more solid at ~20% QoQ in 3Q, but the size of the increase also substantially smaller vs. 2Q as OEMs cut back on handset production & thus memory usage. Most importantly we believe client & enterprise SSD price is rising ~20% QoQ, making the overall NAND price hike close to 20% in 3Q. That however could be still a bit below the latest expectation.

In summary, recent pullback should offer a short-term technical rebound. Jul contracts suggested conventional DRAM & NAND contract prices to rise by close to 20% QoQ in 3QCY26. The size of the increase narrowed notably vs. 2Q, and may be a bit below the latest Street expectation. Shortage should persist into CY27 to keep prices elevated, especially for DRAM, though room for upside is likely limited, given memory is becoming a burden for both AI & non-AI applications & price caps set in some LTAs. That said the recent pullback still opens an opportunity of a near-term rebound. See [Global Memory: Memory becoming a burden of AI too?](#) & [Memory: What do New Memory LTAs mean for sustainability of earnings and multiples? Raising SNDK TP to \\$3000](#) & recent earnings review ([Samsung](#), [SK hynix](#), [Micron](#), [KIOXIA](#), [SNDK](#)) for details.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	6 Aug 2026	Price	TTM	Reported EPS				Reported P/E (x)		
			Closing	Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
005930.KS (SEC- Samsung)	O	KRW	230,000	440,000	201.6%	KRW	6,611.53	48,393	77,273	34.8	4.8	3.0
005935.KS (SEC-Pref - Samsung)	O	KRW	172,100	374,000	165.8%	KRW	6,611.53	48,393	77,273	26.0	3.6	2.2
SMSN.LI (Samsung)	O	USD	4,236.00	7,350.00	199.2%	USD	116.15	812.39	1,290.80	36.5	5.2	3.3
000660.KS (SK hynix)	O	KRW	1,496,000	3,300,000	433.0%	KRW	60,341	395,677	568,862	24.8	3.8	2.6
MU (Micron)	O	USD	881.47	1,300.00	666.4%	USD	8.29	67.39	158.99	106.4	13.1	5.5
285A.JP (KIOXIA)	U	JPY	47,730	40,000	1945.7%	JPY	1,014.00	10,013	9,656.84	47.1	4.8	4.9
SNDK (SanDisk)	O	USD	1,258.58	3,000.00	2868.0%	USD	70.08	238.36	272.09	18.0	5.3	4.6
ASIA			1,909.27									
EM			1,778.30									
SPX			7,709.96									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

MU, SNDK estimate is Adjusted EPS; MU, SNDK valuation is Adjusted P/E (x); SNDK base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Samsung Electronics: We rate Samsung Electronics Outperform with price target of KRW 440,000.

SK hynix : We rate SK hynix Outperform with price target of KRW 3,300,000.

Micron: We rate Micron Outperform with price target of US\$1,300.00.

KIOXIA: We rate KIOXIA Underperform with price target of JPY 40,000.

SanDisk: We rate SNDK Outperform with a price target of \$3,000.00

DETAILS

TrendForce (who publishes research on memory market & also maintains DRAMeXchange for memory spot market) has released the Jul 2026 contract prices for DRAM & NAND covering different applications (dataset can be downloaded at [Memory Price Tracker](#)). We select mainstream products from each segment, apply different weights on them to derive an industry average price change. SSD & HBM are not in the samples but will be additionally considered so that we can compare the overall ASP with our projection. Spot price is noisy but we also summarize some spot price data as it can be a leading indicator of contract price. Please feel free to find our [most recent market review](#) and download our industry and company models from the links below.

- [CoWoS/HBM Model](#)
- [DRAM Industry Model](#)
- [NAND Industry Model](#)
- [Samsung Electronics Model](#)
- [SK hynix Model](#)
- [Micron Model](#)
- [KIOXIA Model](#)
- [SanDisk Model](#)

Jul contracts suggested 17% QoQ price increase for conventional DRAM in 3QCY26. Demand from PC and Mobile softened as expected, but server demand was still strong as customers anticipate more severe shortage next year.

•

Spot price:

- **PC DRAM:** PC DRAM chip spot price continued to be fairly strong in Jul, rising by 8.8% MoM for DDR5 and 17% for DDR4 ([Exhibit 1](#), [Exhibit 3](#), [Exhibit 4](#), [Exhibit 5](#), [Exhibit 6](#)). PC DRAM chip spot price has been remarkably strong, rising by over 30% in the past 3 months. This suggests supply is still very tight in the channel and module makers and distributors are expecting supply shortage to continue or even get worse.
- **Server DRAM:** Server DRAM module spot price was also mostly solid, with DDR4 down 6.7% MoM but DDR5 up another 3.8% after Jun's 26% MoM surge ([Exhibit 1](#), [Exhibit 11](#), [Exhibit 12](#)). Overall server DRAM spot price has also been robust, with DDR5 spot price up 33% in the past 3 months too.

Contract price:

- **Average:** Overall, taking a weighted average of multiple samples, we find contracts set in Jul indicated conventional DRAM prices to rise by 17% QoQ in 3QCY26. This is a tad ahead of our model, but likely below some latest Street expectation. The rate of the increase also narrowed notably vs. 2Q. Please note this is only for conventional DRAM, and the actual company reported ASP may differ due to changing mix among PC, Mobile, server, etc., and also most importantly the balance from steadier HBM price that is largely set by annual contracts ([Exhibit 1](#)). Thus, the weighted average contract price increase here may not match the increase reported in company earnings releases but does serve as an important indication for where the market is heading.
- **PC DRAM:** Contract price in Jul was up by 13-16% MoM, and indicated 3QCY26 contract price to be 17% higher QoQ ([Exhibit 1](#), [Exhibit 3](#), [Exhibit 7](#)). Negotiations are expected to continue beyond Jul but TrendForce expects contract price to stay unchanged in Aug as well. PC shipment is expected to drop by over 10% QoQ in 3QCY26 as OEMs raised PC price to cover memory cost. And as a result OEMs reduced their DRAM purchase, and also became more reluctant towards further

price increases. The pricing power of memory suppliers however remained strong and they have already been diverting production to meet server DRAM demand.

- **PC DDR4:** PC DDR4 prices rose a bit more than DDR5 and DDR5's discount vs DDR4 expanded slightly to 10% in Jul ([Exhibit 5](#), [Exhibit 8](#), [Exhibit 9](#), [Exhibit 10](#)). DDR4 price has been even more robust than DDR5 as supply base largely shifts to Taiwanese makers led by Nanya (2408 TT, not covered). On the other hand demand remains quite resilient from long-tailed customers from consumer segments, such as low-end PC and eSSD.
- **Server DRAM:** Per TrendForce survey, Jul server DRAM contract price was 8-15% higher vs Jun, and suggested 3QCY26 price to be 15% higher than 2QCY26 ([Exhibit 1](#), [Exhibit 11](#), [Exhibit 13](#)). Similarly, negotiations were still ongoing by the end of Jul. Broadly server DRAM demand remained strong as customers foresee supply to get even tighter in 2027, and hence were actively accumulating more inventory. However notably, TrendForce observed that for customers with LTA and price cap in place, Jul price was already near the price ceiling, and also lagged the prices quoted to non-LTA customers. Overall TrendForce projects blended price increase to be 13-18% QoQ for server DRAM in 3QCY26. TrendForce also observes CXMT (688825.CH, not covered)'s price quotes were in line with those from Korean and US suppliers. Additionally, TrendForce does not see the total demand becoming lower, but is observing smaller capacity module getting demand and also enjoying better price increases. TrendForce attributes that to CSPs choosing to pair lower DRAM per CPU core for better memory utilization for Agentic AI. Smaller capacity modules are also easier to source through module makers.
- **Server DDR4:** Server DDR4 contract price rose 15% MoM ([Exhibit 12](#), [Exhibit 14](#)). DDR5's premium over DDR4 narrowed slightly to 15% ([Exhibit 15](#)). Server DDR4 likely will remain in shortage with supply EOL from large suppliers while some legacy applications continue to require it. However, DDR5 was likely already over 90% of the market into 3QCY26 and hence should be the focus.
- **Mobile DRAM:** TrendForce updates Mobile DRAM contract price data only in the first month of a quarter. By the end of Jul, 3QCY26 contract price negotiation was still ongoing, but TrendForce estimates mobile DRAM contract price to increase by ~10% QoQ in 3QCY26, with SK hynix and CXMT potentially seeing more increases over Samsung, as their prices were behind in 2Q ([Exhibit 1](#), [Exhibit 16](#), [Exhibit 17](#), [Exhibit 18](#)). Mobile OEM customers were notably more reluctant than before to accept price hikes, after they meaningfully reduced their smartphone production plans. As a result similar to PC DRAM, mobile DRAM will see the size of price hike notably more moderate in 3Q vs. 2Q. Having said that OEMs were still keen to have supply contracts in place to secure memory supply, at least for the next year. Into 4QCY26, TrendForce expects DRAM demand from smartphone OEMs to remain subdued due to reduced production, and as a result price increase to further narrow to single digit % QoQ.
- **Specialty DRAM:** Specialty DRAM contract price moved up by 12-16% MoM in Jul, and Jul contracts suggested 3QCY26 to be 24-30% higher vs. 2QCY26 ([Exhibit 1](#), [Exhibit 19](#)). After a massive spike in 2QCY26, price increase also had to narrow in 3Q as customers reduce their purchase. Though TrendForce thinks contract price likely will hold up or continue rising for a while as large customers still primarily relied on contract sales from memory vendors, TrendForce also observed spot prices of legacy DRAM falling behind contract price, suggesting demand may be peaking soon.
- **In summary, Jul contracts suggested that 3QCY26 DRAM contract price to continue rising as well, but rate of increase is likely to decelerate meaningfully, and perhaps below some latest expectation. PC and Mobile customers' DRAM inventory were stable or rising, as they had substantially cut their unit production, and as a result were much more reluctant towards price hikes. Strong demand from server customers were unchanged, but price increase was somewhat attenuated by the caps embedded in LTAs. Overall the shortage situation remains and is expected to last into CY27. Price will likely stay elevated, but headroom for further increase may become more limited. See [Global Memory: Memory becoming a burden of AI too?](#) & our earnings reviews ([Samsung](#), [SK hynix](#), [Micron](#) & [KIOXIA](#)) for details.**

NAND wafer contract price appeared to be running out of steam in 3QCY26. Price of Mobile NAND & SSD is still rising but perhaps not as much as expected.

- **Spot price:**

- **NAND Wafer:** Jul NAND wafer spot price was mixed, with 512Gb falling by another 3.1% MoM but 1Tb wafers rebounding by 8.2% MoM ([Exhibit 2](#), [Exhibit 20](#), [Exhibit 21](#)). NAND wafer spot prices had been softening for some time but 1Tb wafer prices showed a notably stronger recovery than 512Gb, perhaps due to more eSSD demand at module makers.

- **Contract price:**

- **Average:** Taking the average of multiple samples of NAND wafer and eMMC/UFS, we find contracts set in Jul indicated just 8% QoQ NAND price increase from 2QCY26 to 3QCY26 ([Exhibit 2](#)). We highlight this is an underestimate due to over representation of wafers and omission of SSDs in our sample. While wafer price is slowing, we believe price hike of SSDs in 3QCY26 likely will around 20% for both client and enterprise, and therefore the blended NAND contract price increase likely should approach 20% as well. And this is mostly in line with our model, though possibly below some latest expectations. Still actual company reported 3QCY26 ASP may differ as product mix changes and timing difference between contract and shipment can have a significant impact on blended ASP.
- **NAND Wafer:** NAND wafer contract price was largely unchanged MoM in Jul. Compared to the average in 2QCY26, Jul contract price was also only mildly higher by 1-4% ([Exhibit 20](#), [Exhibit 21](#), [Exhibit 22](#), [Exhibit 23](#)). TrendForce noted extremely low transaction volume as customers were in complete rejection of any further price hikes. Module makers chose to rely more on their own inventory as they deemed current NAND wafer price too high when compared to sluggish end demand in consumer segment. Going forward, while supply will remain tight, further price increase will likely prove more difficult.
- **eMMC/UFS:** TrendForce expects 3QCY26 Mobile NAND contract price to rise by ~20% QoQ ([Exhibit 24](#), [Exhibit 25](#)). Similar dynamics was in play as in PC & Mobile DRAM. Customers were fighting harder on price increase as they cut production volume & thus memory need. Suppliers still tried to get prices higher to match profitability of eSSD. Overall eMMC/UFS price grew a bit more than other memories mainly to catch up. TrendForce also observed that while global suppliers were more firm on price hikes, Chinese supplier (i.e. YMTC) was not asking as much price increase, and hence won more volume. Going forward, TrendForce expects price hikes will likely become more difficult and moderate in 4QCY26.
- **In summary, NAND wafer price increase is grinding to a halt as module makers stay cautious on softening consumer demand. Mobile NAND price was relatively better but also at a decelerating pace vs. 2Q. Overall 3QCY26 NAND contract price increase likely is somewhat behind latest expectations, even as the price rise of eSSD is better. We think price increase will continue to moderate going forward and perhaps reach a peak some time in CY27. Structurally we remain more cautious on NAND as well, due to more competition from China (details in [our deep dive on YMTC](#)).**

Our industry average is derived from an average of multiple memory products used in different applications, weighted by their contributions to the overall market, and with the weights on DDR5 & LPDDR5 regularly adjusted to reflect its penetration. HBM & SSD are not in our samples & are considered separately.

- Our average DRAM contract price change mentioned above is a weighted average of multiple applications, with Mobile, Server, PC and Specialty DRAM accounting for 34%, 34%, 9% and 15%, respectively. And within PC DRAM, we include both DRAM chips and DRAM module prices in our samples, assuming 10%/90% weight for chips/modules ([Exhibit 26](#)). Additionally, the weights assigned to DDR5 & LPDDR5 will be adjusted according to their adoptions. For NAND, the average is based on a weighted average of eMMC/UFC (40%) and NAND wafer (60%) ([Exhibit 27](#)). We then additionally discuss the

influence of HBM & SSD as they're not in our samples.

EXHIBIT 1: **Jul DRAM contract price indicated 17% QoQ increase for conventional DRAM price in 3QCY26.**

DRAM		Price at End of Month						Quarterly Avg		
		May-26	Jun-26	Jul-26	MoM			2QCY26	3QCY26 QTD	QoQ
					May-26	Jun-26	Jul-26			
Spot Price (US\$/GB)										
PC DRAM	DDR4 8Gb Chip	34.1	36.0	42.1	5.0%	5.6%	17.0%	33.4	40.2	20.3%
	DDR5 16Gb Chip	21.0	23.4	25.5	7.4%	11.8%	8.8%	20.1	24.7	23.0%
	DDR4 64GB RDIMM*	23.1	24.5	22.9	3.1%	6.1%	-6.7%	23.1	23.1	0.1%
	DDR5 64GB RDIMM*	35.5	44.9	46.6	1.6%	26.4%	3.8%	38.1	47.0	23.5%

DRAM		Prior Price			Price for 3QCY26			3QCY26 Over 2QCY26		
		1QCY26	2QCY26	QoQ	Jul-26	Aug-26	Sep-26	Jul-26	Aug-26	Sep-26
Contract Price (US\$/GB, except the price of uMCP & eMCP is module price)										
PC DRAM	DDR4 8Gb Chip	12.5	19.0	52.0%	24.0			26.3%		
	DDR5 16Gb Chip	14.9	18.8	25.7%	22.5			20.0%		
	DDR5 16GB UDIMM*	8.7	12.8	47.8%	14.9			16.1%		
	DDR4 16GB UDIMM*	10.3	14.2	37.6%	16.6			16.7%		
	DDR4 8GB UDIMM*	10.6	14.9	40.0%	17.4			16.8%		
Server DRAM	DDR5 96GB RDIMM*	13.9	21.5	54.9%	24.8			15.7%		
	DDR5 64GB RDIMM*	13.0	20.1	55.5%	23.2			15.2%		
	DDR5 48GB RDIMM*	13.8	21.5	55.5%	24.8			15.2%		
	DDR4 64GB RDIMM*	11.5	17.6	52.7%	20.2			14.9%		
Mobile DRAM	LPDDR5 16GB Discrete	6.3	11.4	80.6%		12.7			10.7%	
	LPDDR5 12GB Discrete	6.3	11.4	80.2%		12.7			10.7%	
	LPDDR4X 8GB Discrete	6.2	10.9	76.9%		12.2			11.9%	
	LPDDR4X 4GB Discrete	6.2	10.9	77.4%		12.3			11.9%	
	uMCP (LPDDR4X 64Gb+NAND 128GB)**	65.7	119.4	81.9%		131.5			10.1%	
	eMCP (LPDDR4X 32Gb+NAND 64GB)**	34.8	76.6	120.0%		78.4			2.3%	
Specialty DRAM	DDR4 4Gb	12.7	23.3	84.2%	29.0			24.3%		
	DDR3 4Gb	12.0	22.3	86.1%	29.0			29.9%		
Weighted Average		10.1	16.7	65.2%	19.6			17.0%		
Bernstein Estimate (Blended ASP, inc HBM)****										
Samsung		9.5	14.9	56.7%		16.5			10.3%	
SK hynix		8.3	12.5	50.7%		13.9			11.1%	
Micron***		7.1	10.7	50.4%		12.3			15.4%	
Weighted Average		8.5	13.0	53.2%		14.5			11.6%	

*: DIMM = Dual In-line Memory Module; UDIMM (Unbuffered DIMM) is the DIMM mostly for desktop PC; RDIMM (Registered DIMM) is mostly for server

** eMCP (Embedded Multi-Chip Package) is mostly used in mobile applications & is a package standard that extends the eMMC standard (which consists of NAND & controller) to include DRAM too in the package. uMCP (UFS + MCP) is similar to eMCP but replace eMMC with UFS. Their prices include the DRAM, NAND & controller in the package but are classified as part of DRAM statistics as DRAM contributes more value.

***: Micron's quarter ends a month earlier.

****: Our models were last updated in Jun 2026.

Source: DRAmExchange, Bernstein estimates and analysis

EXHIBIT 2: Our sample for NAND only includes wafer and eMMC/UFS, which together rose by 8.4% QoQ in Jul. However we understand SSD contract price likely is also rising growing 20% QoQ, and with that included overall NAND contract price likely increased can also rise by close to 20% QoQ in 3QCY26.

NAND		Price at End of Month						Quarterly Avg		
		May-26	Jun-26	Jul-26	MoM			2QCY26	3QCY26	QoQ
					May-26	Jun-26	Jul-26			
Spot Price (US\$/GB)										
NAND Wafer	1Tb 3D TLC/QLC	0.197	0.190	0.206	0.2%	-3.3%	8.2%	0.197	0.202	2.5%
	512Gb 3D TLC	0.323	0.310	0.301	0.2%	-3.8%	-3.1%	0.322	0.302	-6.2%

NAND		Prior Price			Price for 3QCY26			3QCY26 Over 2QCY26		
		1QCY26	2QCY26	QoQ	Jul-26	Aug-26	Sep-26	Jul-26	Aug-26	Sep-26
Contract Price (US\$/GB)										
NAND Wafer	1Tb 3D QLC	0.191	0.235	23.1%	0.237			0.8%		
	1Tb 3D TLC	0.197	0.246	24.5%	0.254			3.3%		
	512Gb 3D TLC	0.312	0.396	27.2%	0.403			1.6%		
eMMC/UFS*	512GB UFS	0.132	0.238	80.0%		0.284			19.1%	
	256GB UFS	0.131	0.236	80.0%		0.286			21.3%	
	128GB TLC eMMC	0.131	0.231	75.4%		0.280			21.2%	
Weighted Average		0.193	0.269	39.9%	0.292			8.4%		
Bernstein Estimate (Blended ASP, inc eSSD)****										
Samsung		0.156	0.259	66.0%		0.316			22.0%	
	SK hynix	0.168	0.279	66.0%		0.341			22.0%	
	Micron**	0.141	0.232	65.0%		0.302			30.0%	
	KIOXIA	0.107	0.178	66.0%		0.217			22.0%	
	SanDisk	0.242	0.314	29.9%		0.343			9.0%	
Weighted Average		0.155	0.248	59.3%	0.300			21.1%		

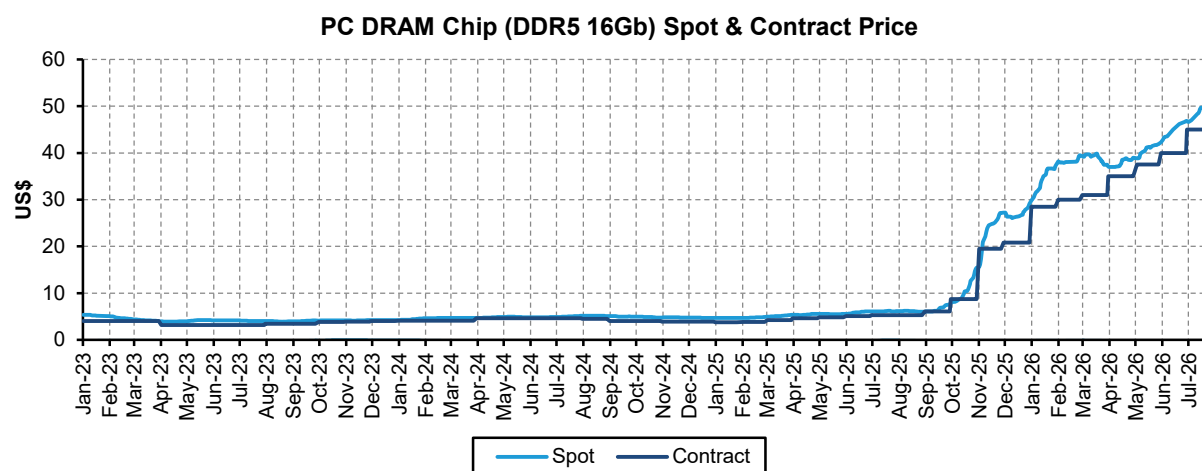
*: eMMC (Embedded Multi Media Card) is a standard that packages NAND dies & controller together. UFS (Universal Flash Storage) is an enhancement over eMMC.

** : Micron's quarter ends a month earlier.

****: Our models were last updated in Jun 2026.

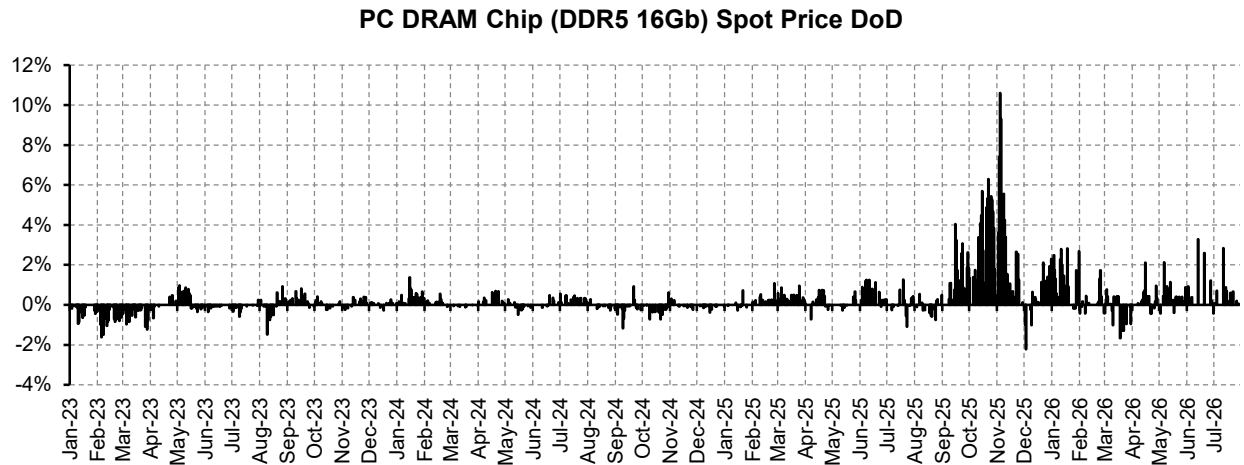
Source: DRAmEXchange, Bloomberg, Bernstein estimates and analysis

EXHIBIT 3: PC DDR5 DRAM chip spot price continued rallying in Jul.



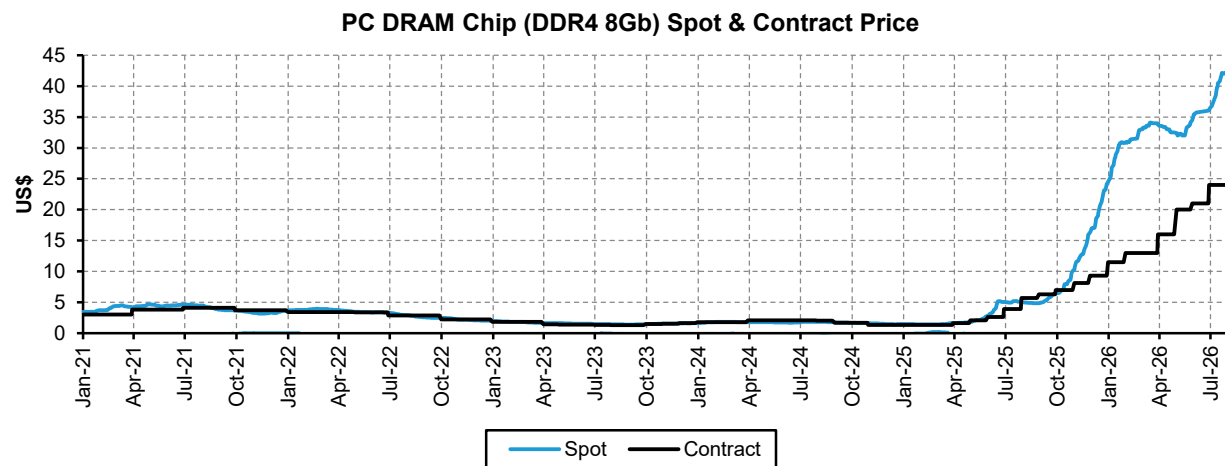
Source: DRAmEXchange and Bernstein analysis

EXHIBIT 4: **PC DDR5 chip spot price saw some correction in late Mar, but it has since been growing almost non-stop.**



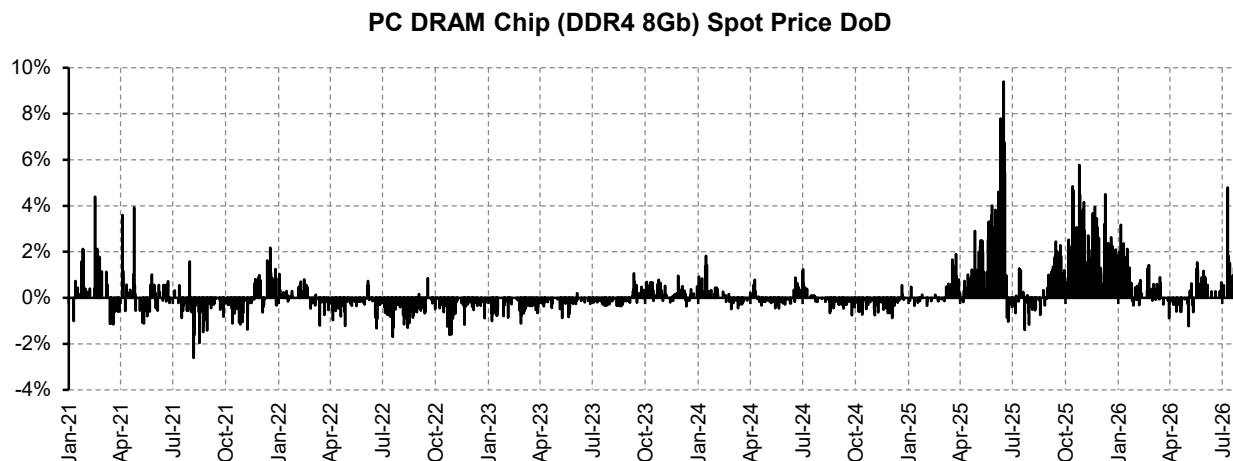
Source: DRAmEXchange & Bernstein analysis

EXHIBIT 5: **PC DDR4 chip spot price also increased meaningfully in Jul.**



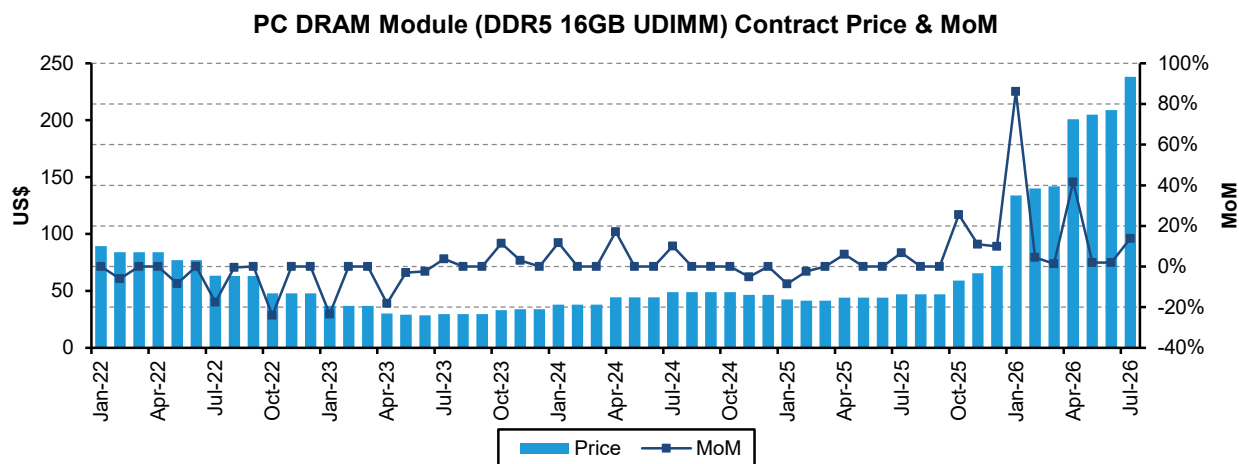
Source: DRAmEXchange & Bernstein analysis

EXHIBIT 6: **Jul saw some notable spike for daily PC DDR4 spot price increase.**

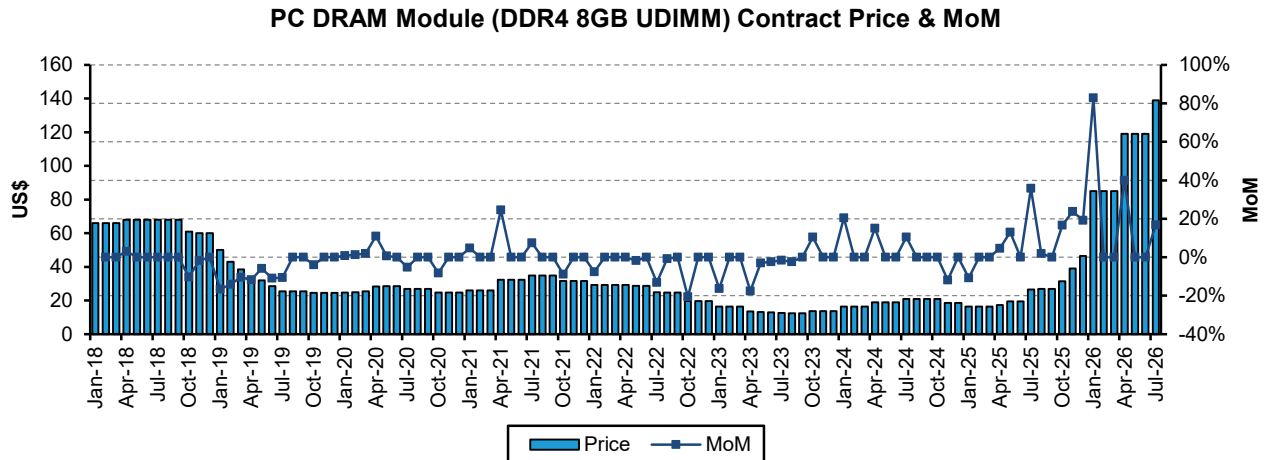


Source: DRAmExchange & Bernstein analysis

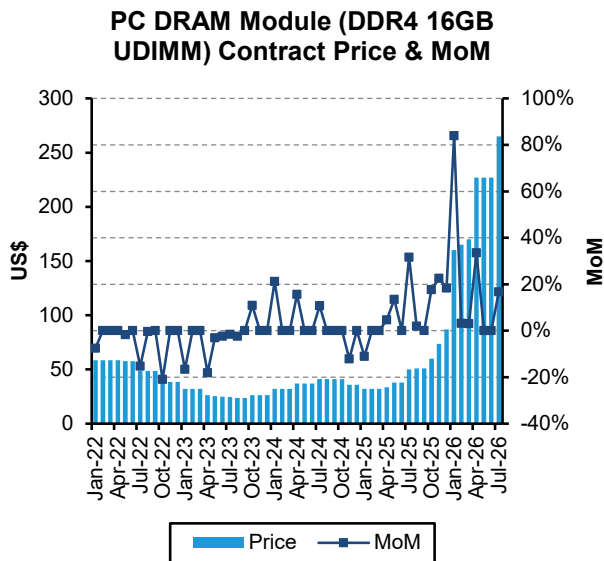
EXHIBIT 7: **PC DDR5 module contract price increased by 14% MoM in Jul as OEMs negotiate for 3QCY26 contracts.**



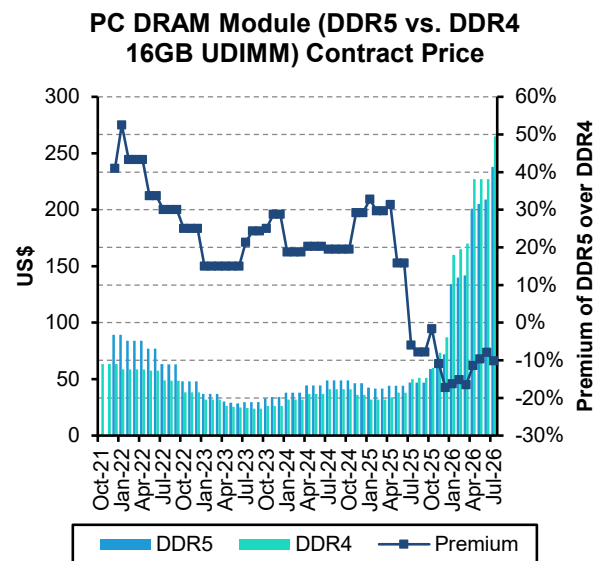
Source: DRAmExchange & Bernstein analysis

EXHIBIT 8: **PC DDR4 module contract price was up by 17% MoM in Jul.**

Source: DRAmExchange & Bernstein analysis

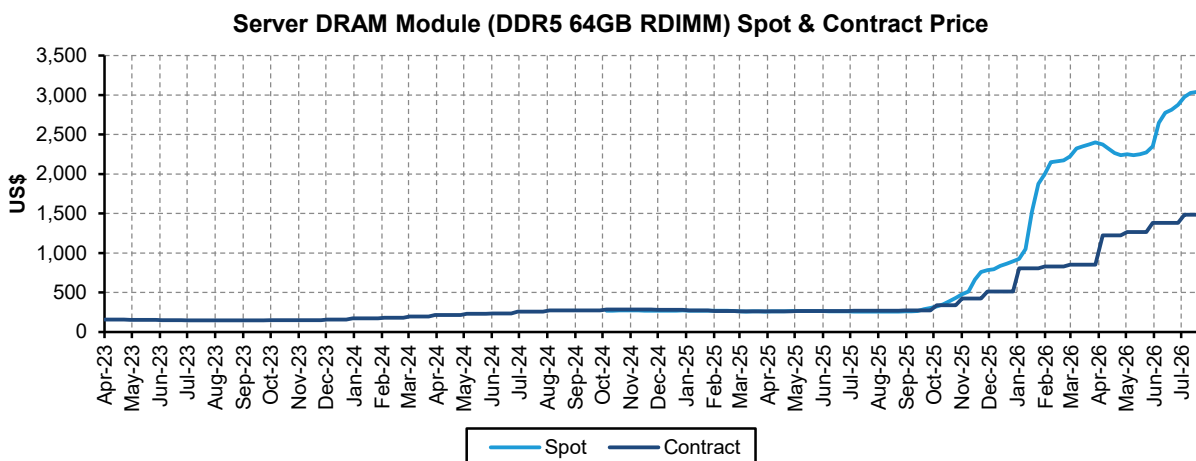
EXHIBIT 9: **16GB DDR4 modules contract price grew at similar rate to 8GB.**

Source: DRAmExchange & Bernstein analysis

EXHIBIT 10: **PC DDR5 price discount vs. DDR4 was at 10%.**

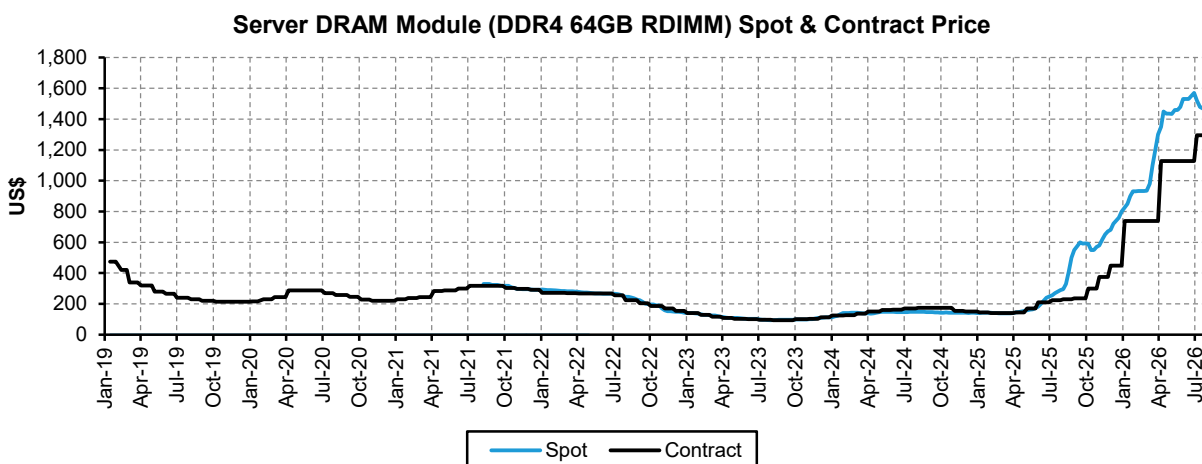
Source: DRAmExchange & Bernstein analysis

EXHIBIT 11: **Server DDR5 module spot price further grew in Jul after a notable rise in Jun.**



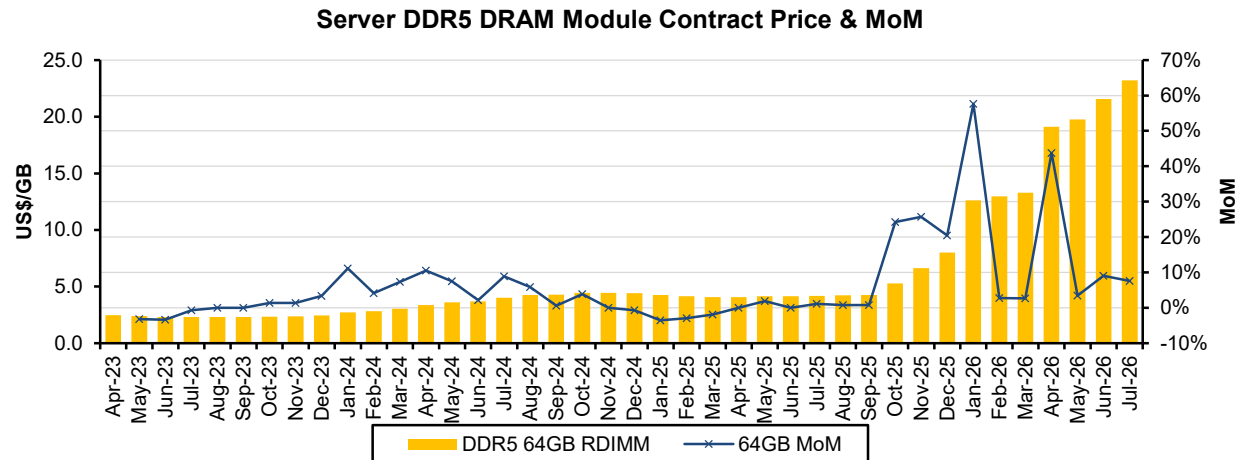
Source: DRAmExchange & Bernstein analysis

EXHIBIT 12: **Server DDR4 module spot price however had some mild correction in Jul.**

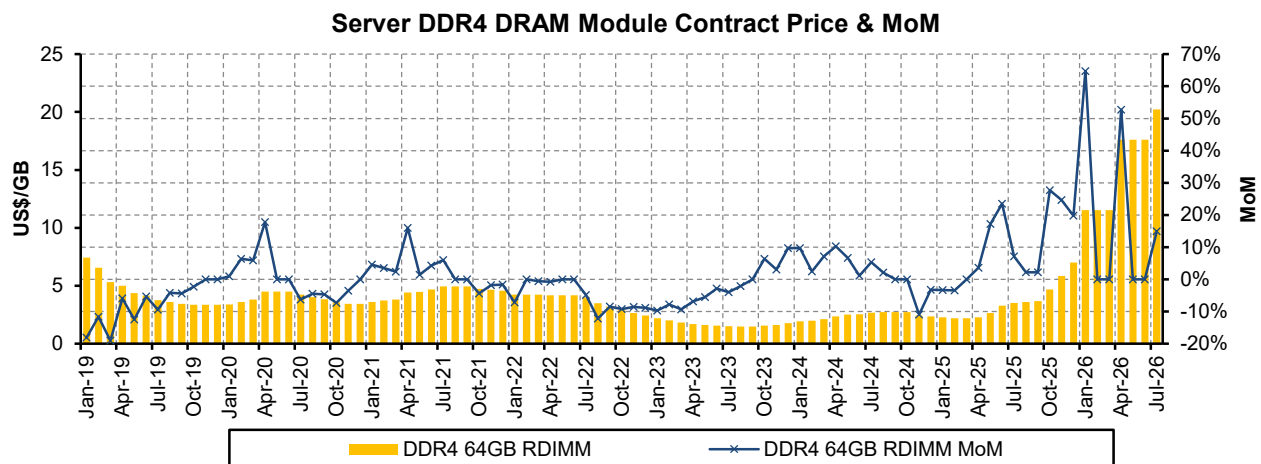


DRAmExchange started to track server spot price in Aug 2021 as its trading volume became meaningful.

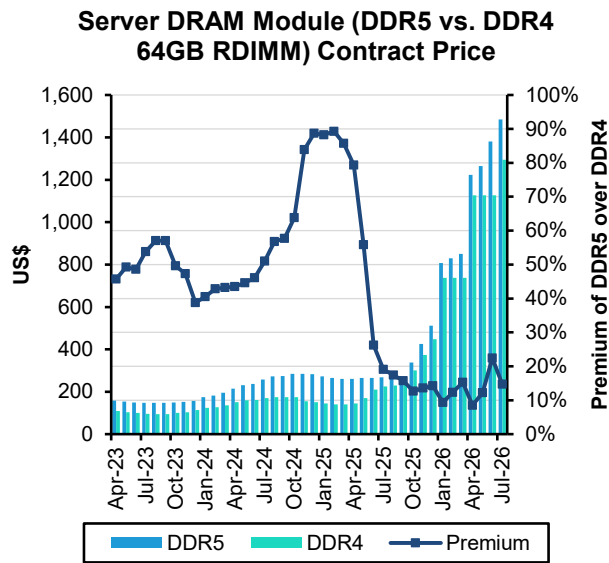
Source: DRAmExchange & Bernstein analysis

EXHIBIT 13: **Jul server DDR5 module contract price rose by 7.6% MoM.**

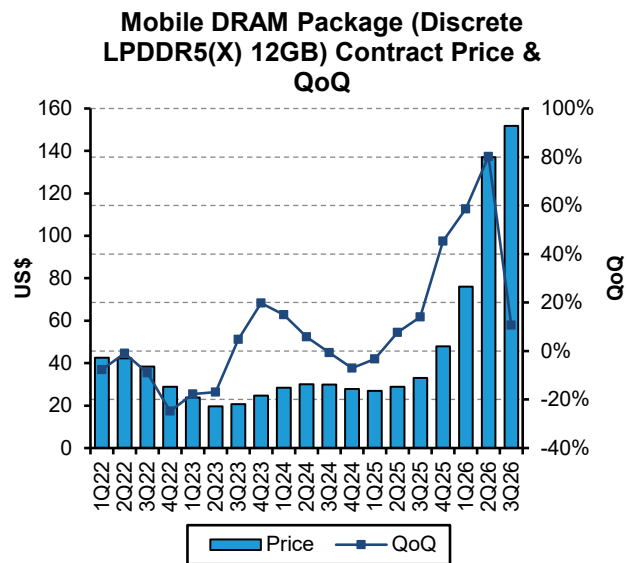
Source: DRAmExchange and Bernstein analysis

EXHIBIT 14: **Server DDR4 module contract price went up more by 15% MoM.**

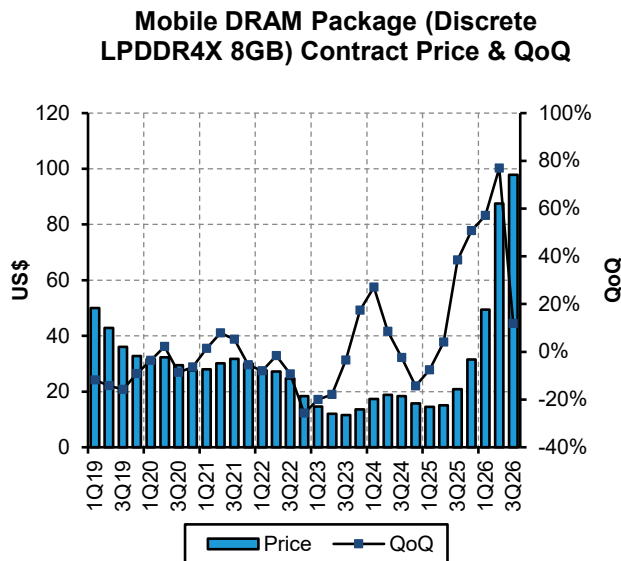
Source: DRAmExchange & Bernstein analysis

EXHIBIT 15: Server DDR5 premium over DDR4 narrowed to 15% in Jul.

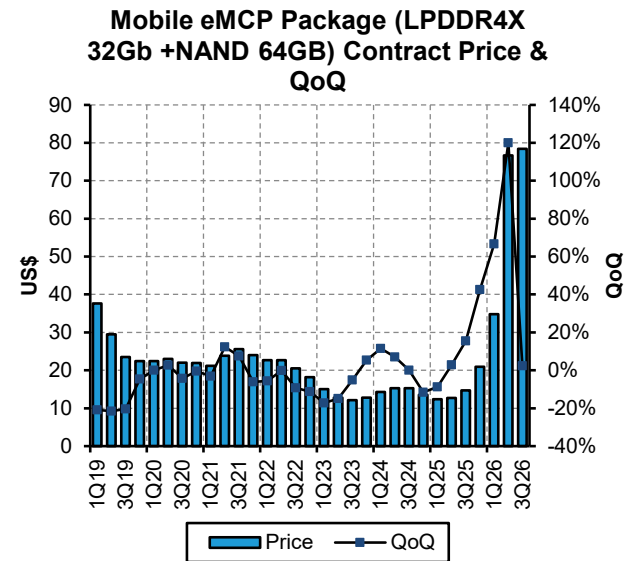
Source: DRAmExchange and Bernstein analysis

EXHIBIT 16: LPDDR5 contract price was estimated to rise by only 11% in 3QCY26.

Source: DRAmExchange and Bernstein analysis

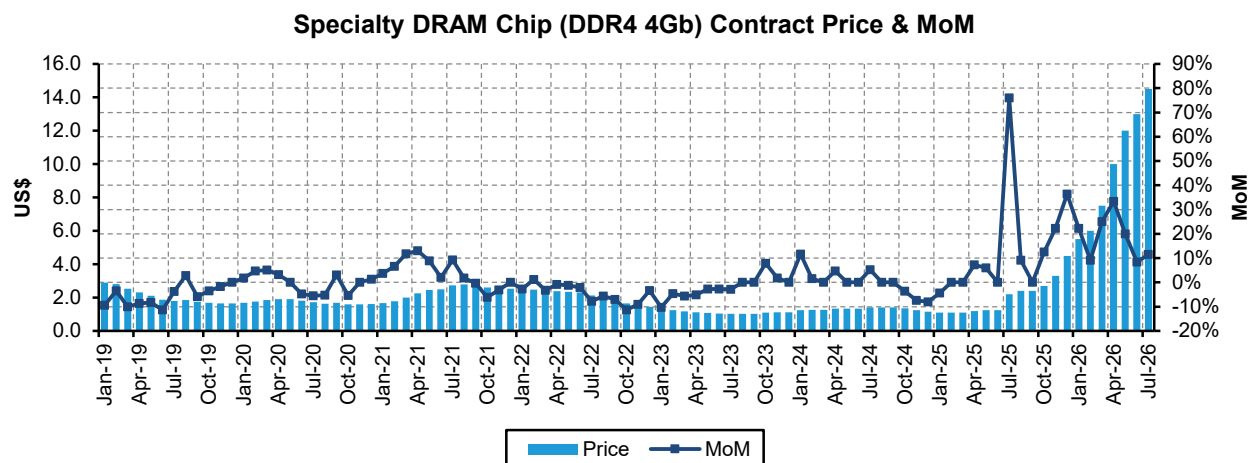
EXHIBIT 17: LPDDR4 price also estimated to rise by just 12% QoQ.

Source: DRAmExchange and Bernstein analysis

EXHIBIT 18: The price of low-end eMCP packages was largely flattish at 2.3% up QoQ for 3QCY26.

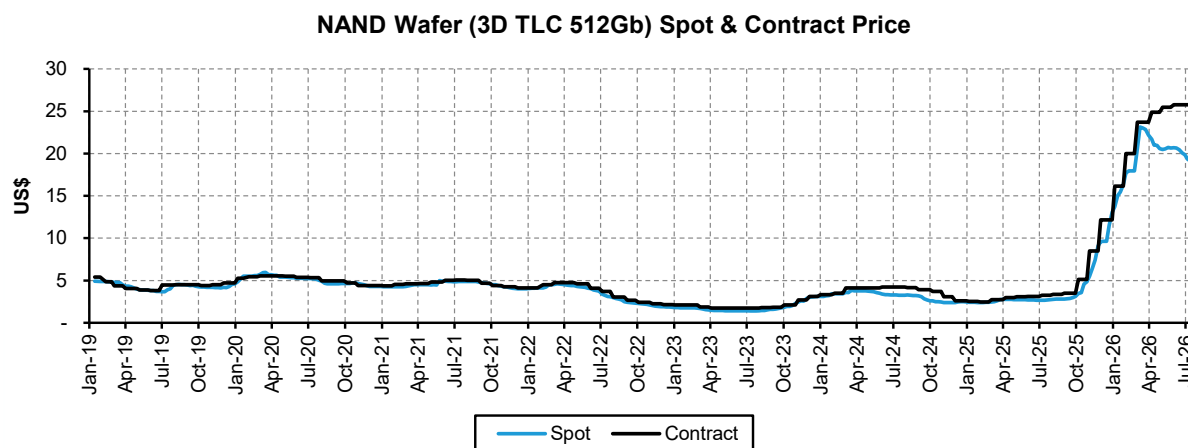
Source: DRAmExchange & Bernstein analysis

EXHIBIT 19: **Consumer DRAM contract price increase remained more robust, rising by another ~14% MoM in Jul and indicated 27% QoQ rise for 3QCY26.**

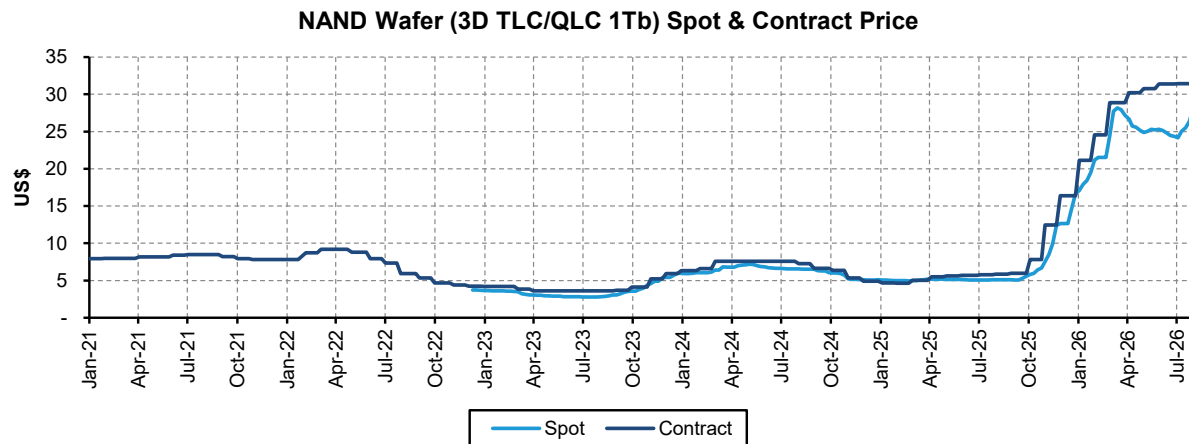


Source: DRAmEXchange & Bernstein analysis

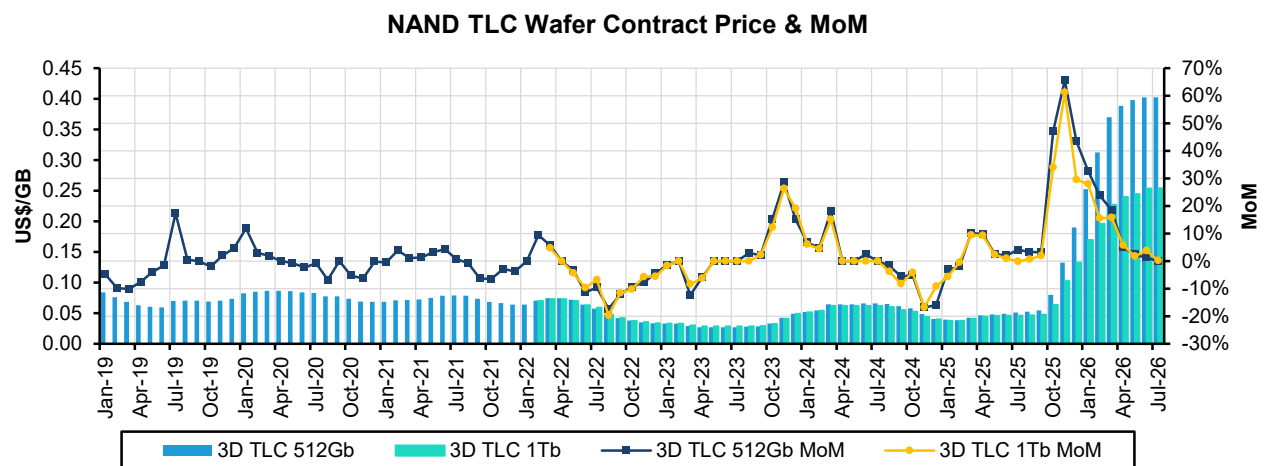
EXHIBIT 20: **NAND 512Gb spot price remained soft in Jul.**



Source: DRAmEXchange & Bernstein analysis

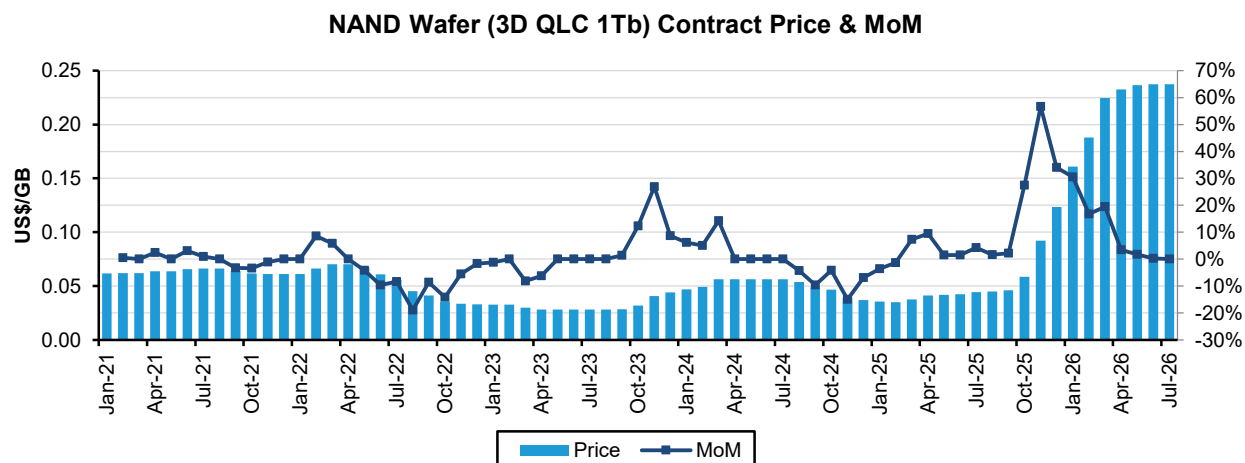
EXHIBIT 21: **1Tb wafer spot price also decreased.**

Source: DRAmEXchange and Bernstein analysis

EXHIBIT 22: **TLC NAND wafer contract price largely unchanged MoM in Jul as module makers reject price hikes on demand concerns.**

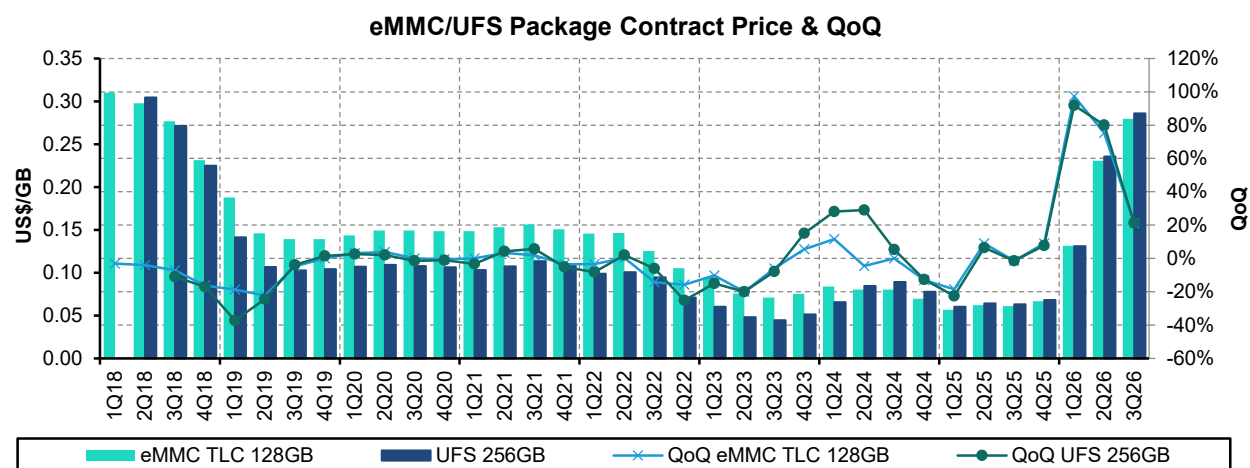
Source: DRAmEXchange & Bernstein analysis

EXHIBIT 23: **QLC NAND wafer contract price was also flattish.**



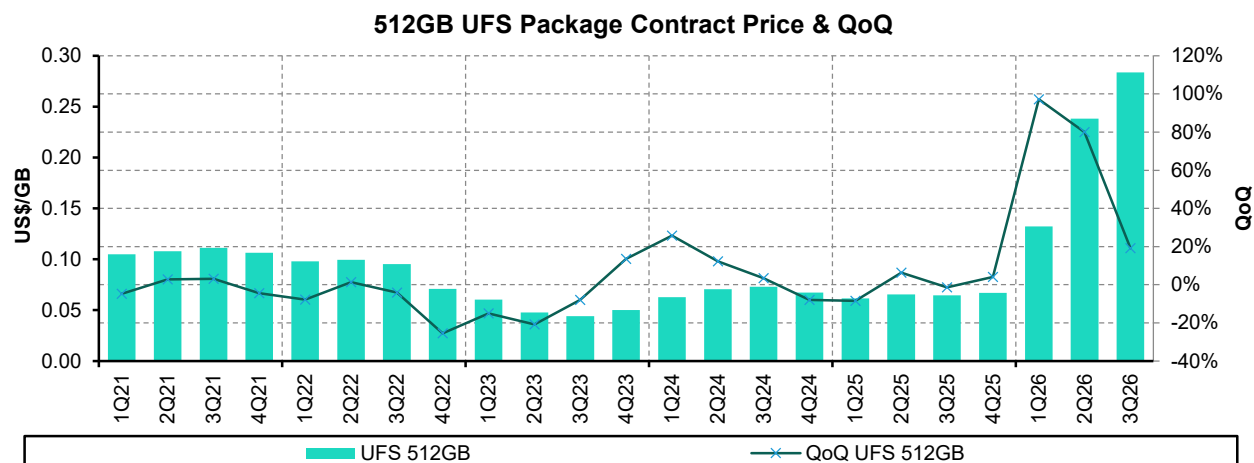
Source: DRAmExchange & Bernstein analysis

EXHIBIT 24: **Mobile NAND contract price for 3QCY26 is estimated to increase by 20% QoQ, notably slower vs 2Q as customers cut demand and push ed back on price hikes.**



Source: DRAmExchange & Bernstein analysis

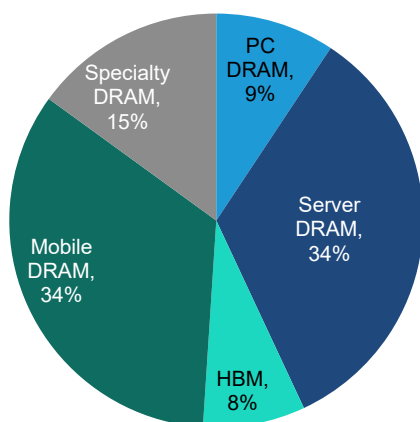
EXHIBIT 25: **High-density 512GB UFS contract price is also expected to rise by ~20% in 3QCY26.**



Source: DRAmExchange and Bernstein analysis

EXHIBIT 26: **Our industry ASP is a weighted average of different products based on the size of their demand. We then further adjust it for HBM.**

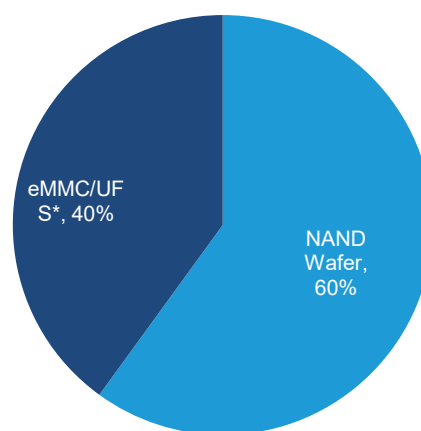
DRAM ASP Weight by Products



Source: Bernstein analysis

EXHIBIT 27: **We allocate 40% & 60% to eMMC/UFS & wafer, respectively, to derive the industry NAND ASP. eSSD is then considered in addition.**

NAND ASP Weight by Products



Source: Bernstein analysis

Further Reading Materials

- 3 Aug 2026 - [2Q26 AI Server Pulse: Summertime Cool-Down](#)
- 31 Jul 2026 - [KIOXIA 1QFY26: Can buyback help KIOXIA avoid volatility?](#)
- 30 Jul 2026 - [Samsung 2Q26: Poised to Lead in HBM in 2H26?](#)
- 29 Jul 2026 - [SK hynix 2Q26: ASP should improve in 2H26. Overall thesis intact](#)
- 27 Jul 2026 - [Global Memory, NVIDIA & Broadcom: Quick thoughts on strategic partnerships](#)

- 25 Jun 2026 - [Micron 3QFY26: Strong near-term momentum but SCAs thus far still can't avoid cyclical](#)
- 20 Jul 2026 - [Korea Memory Export Tracker \(Jun\): Strong HBM in 2Q26, esp. for Samsung](#)
- 20 Jul 2026 - [Global Memory: New Memory LTAs Part 3 - Bull vs Bear Debate - Webinar slides, transcript and summary](#)
- 30 June 2026 - [Memory: What do New Memory LTAs mean for sustainability of earnings and multiples? Raising SNDK TP to \\$3000](#)
- 22 June 2026 - [Global Memory: Memory becoming a burden of AI too?](#)
- 15 June 2026 - [KIOXIA: All parties come to an end - Underperform](#)
- 2 June 2026 - [KIOXIA Investor Day: 22% bit CAGR good enough?](#)
- 1 May 2026 - [Sandisk FQ3'26: Nosebleed ASPs \(now locked in\); Raise estimates and TP to \\$1700, Reiterate Outperform \(top ST pick\)](#)
- 13 Apr 2026 - [Asia Semiconductors & Global Memory: AI, war, memory price - 1Q26 earnings preview](#)
- 9 Apr 2026 - [Sandisk: How high can estimates and the stock go? Raising estimates and TP to \\$1,250, with blue-sky scenario at \\$3000](#)
- 17 Mar 2026 - [Asia Semiconductors & Hardware: Stronger & longer - 2026 Tech Tour takeaways](#)
- 13 Mar 2026 - [Global Industrial Gases & Semis: Assessing the helium impact](#)
- 27 Jan 2026 - [AI Value Chain: Putting it all together - how much AI upside does each name really have, and who might be an underappreciated winner?](#)
- 2 Jan 2026 - [Asia Semiconductors & Global Memory: 2026 is still all about AI](#)
- 11 Dec 2025 - [Expert Insight: How much memory do AI Data Centers need?](#)
- 24 Nov 2025 - [Asian Semis & Global Memory: Two findings, one simulation & one model for AI](#)
- 27 Oct 2025 - [AI Value Chain: How much does a GW of data center capacity actually cost, and what goes into it?](#)
- 1 Sep 2025 - [Global Semicap and Memory - US revokes Samsung and SK hynix semicap license waivers for their China Fabs](#)
- 21 Aug 2025 - [KIOXIA: Decipher the complex JV with SanDisk](#)
- 29 Jul 2025 - [Samsung: Dimensioning the contribution of the Tesla project on Samsung & ASML](#)
- 13 Jun 2025 - [KIOXIA: The Last Samurai - initiating with Underperform](#)
- 6 Dec 2024 - [Global Memory: Decoupling in memory? A quantitative assessment](#)
- 21 Nov 2024 - [Global Memory: Will China/YMTC surprise us in NAND?](#)
- 16 Sep 2024 - [Global Memory: Is Samsung still the memory leader?](#)
- 8 Jul 2024 - [Global Semiconductors: AI semiconductor roadmap - 3 charts & 7 investment implications](#)
- 1 Jun 2023 - [10 FAQs on Investing in Semiconductors for Generative AI](#)
- 4 Mar 2022 - [Global Memory: Cornerstone of the Data Economy](#)

- 25 Oct 2021 [Global Memory: The cornerstone of the data economy but painful cyclical - Initiating with Underperform](#)

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**Samsung Electronics Co Ltd**

We value Samsung Electronics with 6.2x of forward 5Q-8Q EPS of KRW71,172 and arrive at price target of KRW 440,000 for its common shares, KRW374,000 for preferred shares, and US\$7,350 for GDR.

SK Hynix Inc

We value SK hynix with 6.2x of forward 5Q-8Q EPS of KRW536,158 and arrive at price target of KRW3,300,000.

Micron Technology Inc

We value Micron with 7.7x of forward 5Q-8Q EPS of US\$169.8 and arrive at price target of US\$1,300.

KIOXIA Holdings Corp

We value KIOXIA with 4.1x of 1-year forward EPS of ¥9,810 and arrive at price target of ¥40,000.

SanDisk Corp

We value SNDK at \$3000 on 11x FY28 EPS, or 14x through cycle EPS (FY26-30 avg EPS).

RISKS**Samsung Electronics Co Ltd**

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

SK Hynix Inc

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

Micron Technology Inc

The biggest downside risk to our target price is an earlier end of favorable pricing environment, which can be a result of weaker demand or higher supply. Investor sentiment and hence the valuation rewarded by investors is also a risk. China's progress in memory, especially in NAND, is a downside risk too.

KIOXIA Holdings Corp

Upside risk to our target price include 1) stronger than expected NAND demand growth, which likely would lead to better NAND pricing and hence company earnings, 2) favorable policy support from Japanese government such as capex subsidies, and 3) better than expected cost reduction

SanDisk Corp

The biggest risks to the downside on Sandisk and to our price target are that: 1) Near-term numbers look high and face cyclical downside from NAND; 2) SNKD's disclosures and broader investor communications have been confusing, potentially dissuading more quality-focused investors; and 3) NAND weakness could extend beyond the current cycle and be more structural, in which case Sandisk's DCF value could be structurally lower and the impaired value of its assets may be materially below replacement cost.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp

- Neutral (N): Stock will perform in line with the market index to within +/-10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Neutral (C-N): The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Underperform (C-UP): The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side)

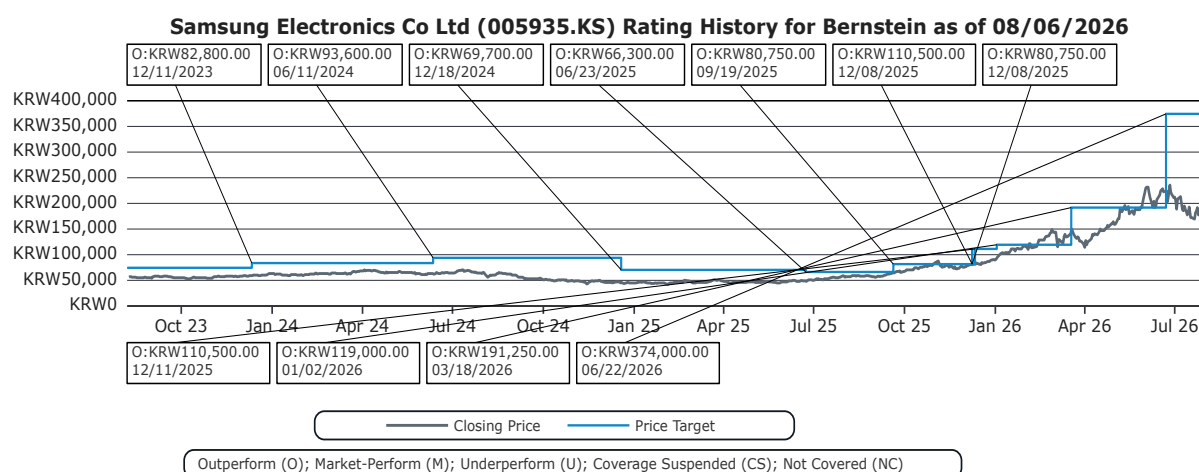
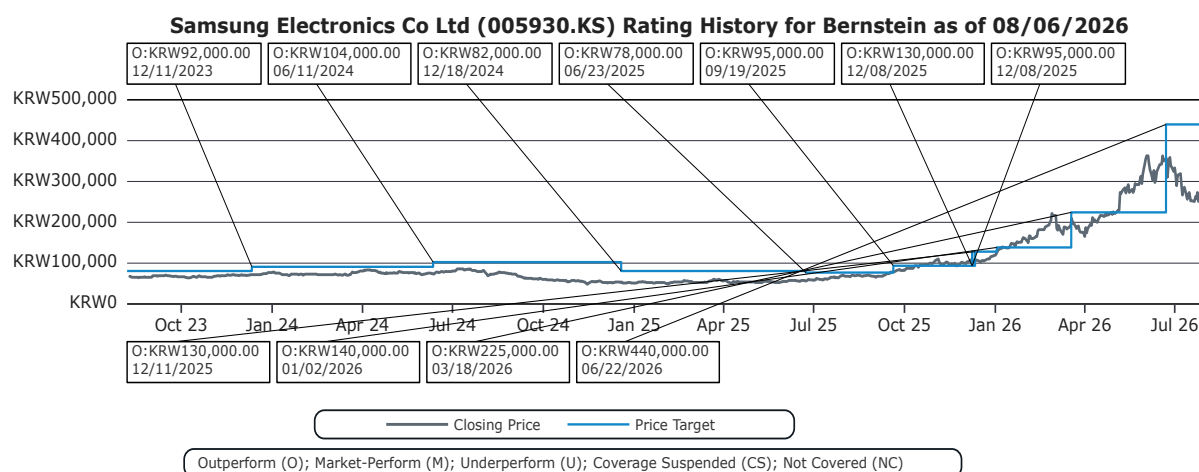
within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

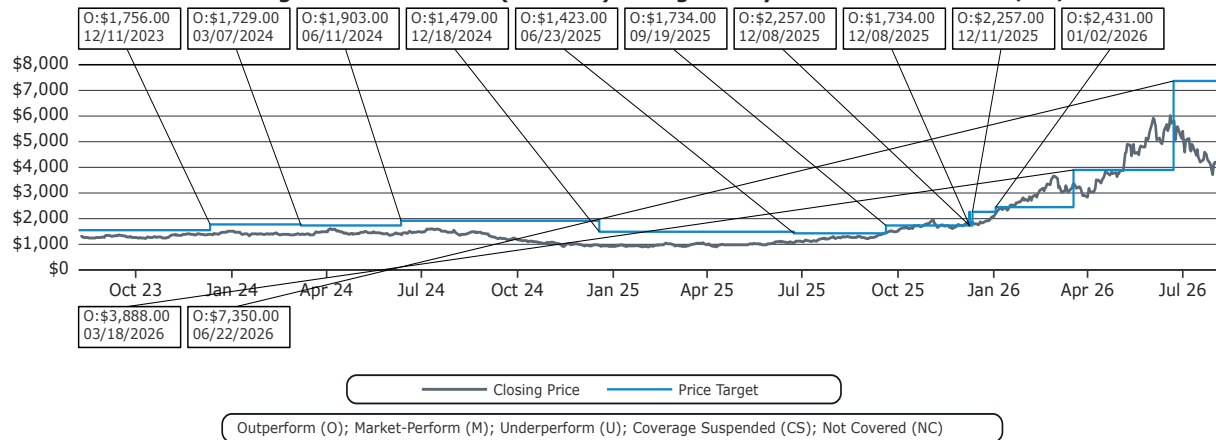
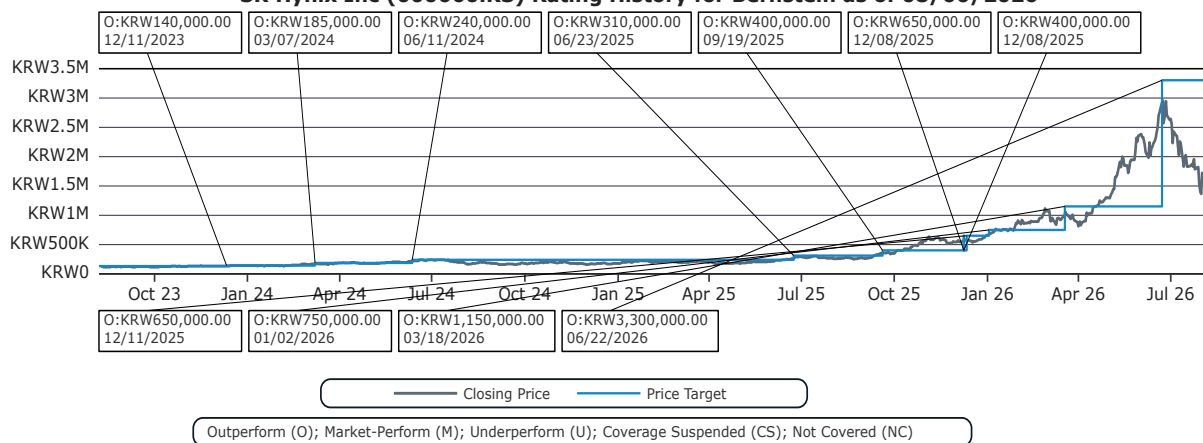
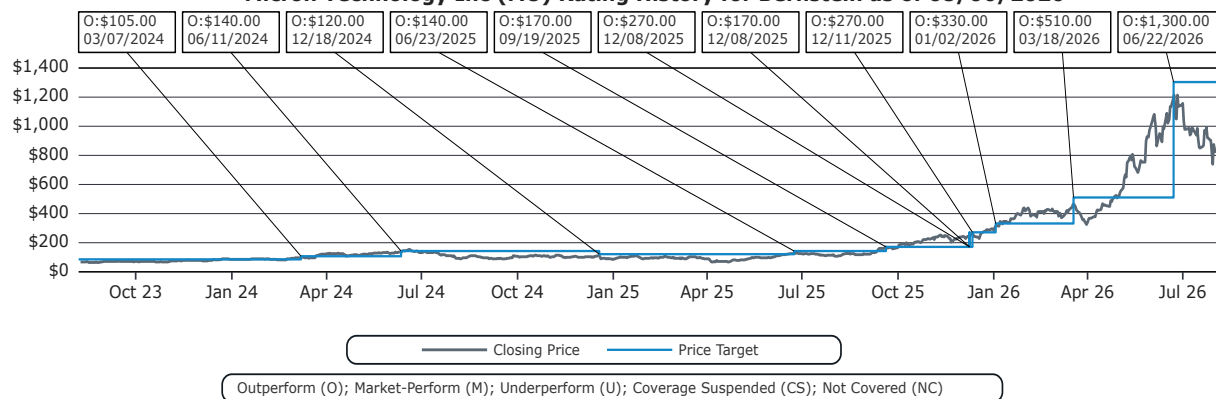
DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

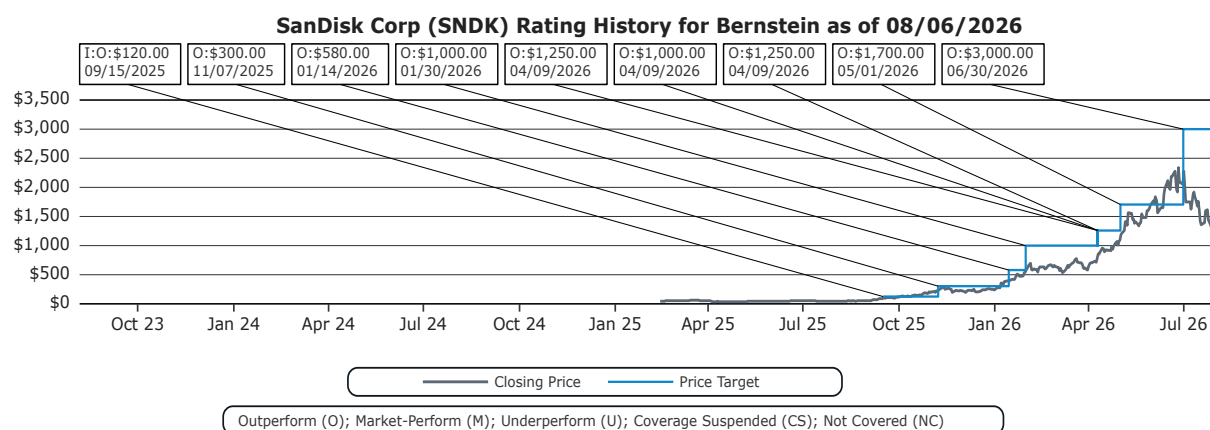
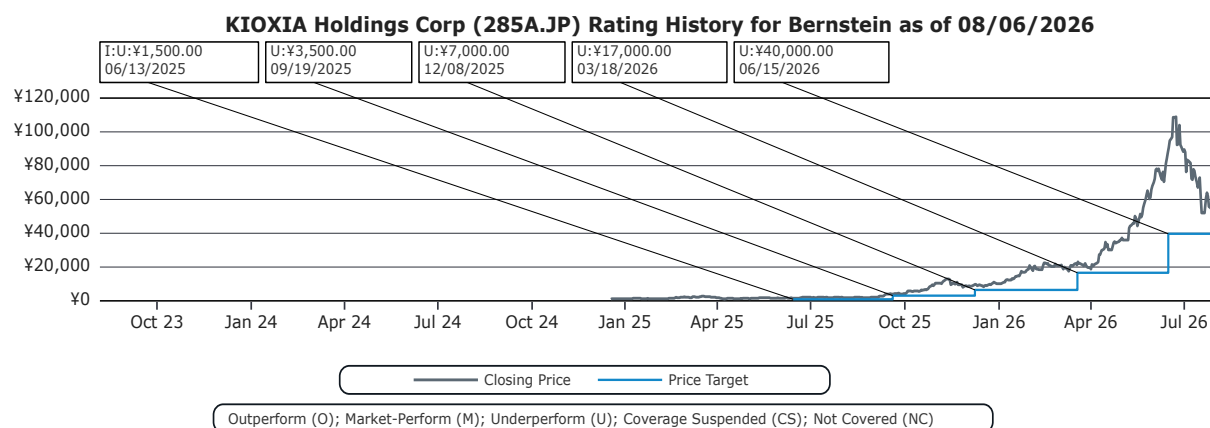
Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY



Samsung Electronics Co Ltd (SMSN.LI) Rating History for Bernstein as of 08/06/2026**SK Hynix Inc (000660.KS) Rating History for Bernstein as of 08/06/2026****Micron Technology Inc (MU) Rating History for Bernstein as of 08/06/2026**



All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

CONFLICTS OF INTEREST

SG and/or its affiliates beneficially own 0.5% or more of [the total issued share capital/any class of common equity securities] with a net [long/short] position of: Micron Technology Inc and KIOXIA Holdings Corp.

SG and/or its affiliates beneficially own 1% or more of a class of common equity securities of the following company: KIOXIA Holdings Corp.

Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Samsung Electronics Co Ltd.

Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Samsung Electronics Co Ltd, SK Hynix Inc and KIOXIA Holdings Corp.

Certain affiliates of Bernstein act as market maker or liquidity provider in the debt securities of: SK Hynix Inc and KIOXIA Holdings Corp.

Certain affiliates of Bernstein act as market maker or liquidity provider in the equities securities of: Micron Technology Inc.

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

- +1 Bernstein Institutional Services LLC; New York, New York, USA
- +44 Bernstein Autonomous LLP; London UK
- +212 Société Générale Africa Technologies & Services; Casablanca, Morocco
- +33 BSG France S.A.; Paris, France
- +34 BSG France S.A.; Madrid, Spain
- +41 Bernstein Autonomous LLP; Geneva, Switzerland
- +49 BSG France S.A.; Frankfurt, Germany
- +91 Sanford C. Bernstein (India) Private Limited; Mumbai, India
- +852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China
- +65 Sanford C. Bernstein (Singapore) Private Limited; Singapore
- +81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side) within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for "Bernstein" and "Autonomous" research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the "Autonomous" and "Bernstein" brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.
- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to "Autonomous" branded products (including certain Sales materials) please visit:

www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation ("MAR") and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission ("SEC") and a member of the U.S. Financial Industry Regulatory Authority, Inc. ("FINRA") is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as "permitted client" as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. ("BTG") is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Financial Promotion Order"), (ii) are persons falling within Article 49(2)(a) to (d) ("high net worth companies, unincorporated associations, etc.") of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as "relevant persons"). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (<https://www.sfc.hk/publicregWeb/corp/AXC846/details>). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore ("SFA"). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration

No. 20213710W and located at 8 Marina Boulevard, #12-01, Marina Bay Financial Centre, Singapore 018981, +65-6326-7000.

To our readers in the People's Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People's Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the "PRC") in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Investment Management Association of Japan. This publication is solely for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. ("Daiwa"), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;
- dealing in a financial product;
- making a market for a financial product; and
- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).
- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com.
- SCB India does not have any disciplinary history as on the date of this report.
- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives

- do not have any financial interest in the subject company
- do not have actual/beneficial ownership of one percent or more in securities of the subject company;
- is not engaged in any investment banking activities for Indian companies, as such;
- have not managed or co-managed a public offering in the past twelve months for any Indian companies;
- have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;
- have not received compensation for brokerage services from the subject company in the past twelve months;
- have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and
- do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.
- do not have any conflict of interest in the subject company as of the date of this report.
- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.
- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.
- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com
- The Research investor charter and Terms & Conditions of SCB India are available on its website and may be accessed at [Sanford C. Bernstein \(India\) Private Limited](https://bernsteinresearch.in/) (<https://bernsteinresearch.in/>) for your reference.
- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act ("CISA") and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park

Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are "eligible counterparties", "professional clients", "institutional investors" and/or "professional investors" as defined by the aforementioned regulators, and must not be redistributed to retail clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein's express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein's express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein's research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)' current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or

subscribe for shares, or to induce engagement in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients' home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg"). Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg's licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.

No part of this material may be reproduced, distributed or transmitted or otherwise made available without prior consent of the entities noted herein. Copyright Bernstein Institutional Services LLC Bernstein Autonomous LLP, BSG France S.A., Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited and Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社). All rights reserved. The trademarks and service marks contained herein are the property of their respective owners. Any unauthorized use or disclosure is strictly prohibited. The entities noted herein may pursue legal action if the unauthorized use results in any defamation and/or reputational risk to the entities noted herein and research published under the Bernstein and Autonomous brands.